

Ievgeniia (Ivy) Boiko

iboiko@carnegiescience.edu — ivyvb18@gmail.com — ivyboiko.github.io

EDUCATION

California State University, Long Beach - Long Beach, CA **2023 - 2025**
BACHELOR OF SCIENCE IN PHYSICS

Monterey Peninsula College - Monterey, CA **2021 - 2023**

RESEARCH EXPERIENCE

Carnegie Observatories - Pasadena, CA **2026 - present**
PROJECT RESEARCH ASSISTANT; projects include:

- COMPUTATIONAL ASTRONOMY; re-analyzing high-quality GALAH red giant branch spectra to measure rubidium abundances for a large stellar sample by fitting synthetic stellar spectra using MOOG line analysis code and MCMC-based fitting tools to place Rb in context with other neutron-capture elements already measured by GALAH in order to create a value-added catalogue

California State University, Long Beach - Long Beach, CA **2024 - 2025**
UNDERGRADUATE RESEARCH ASSISTANT; projects included:

- COMPUTATIONAL ASTRONOMY; studying the chemical bimodality in the Milky Way through analysis of a spectroscopic-asteroseismic catalogue (APO-K2) with comparison to a large-scale cosmological simulation (NIHAO); worked toward determining stellar ages with use of an asteroseismology pipeline (BAM)

Monterey Institute for Research in Astronomy - Marina, CA **2022 - 2024**
VOLUNTEER AND INTERN; projects included:

- OBSERVATIONAL ASTRONOMY; operated and collected data on the polarization of various stellar and planetary targets with both a 14 and 36-inch telescope, contributed data obtained from independently-conducted observations to multiple publications
- SOFTWARE INSTALLATION, DEBUGGING, AND READING/WRITING PROGRAMS; installed and used a Fortran program on a Linux system to analyze data in order to determine the D/H ratio in the atmosphere of Neptune
- WRITING; made use of online databases such as ADS and SIMBAD to conduct a literature review to contribute to a research paper on high polarization standard stars

PUBLICATIONS

Published:

1. Jeremy Bailey, Daniel V. Cotton, Kimberly Bott and **Ievgeniia Boiko**, *One hundred years of Venus polarimetry: PICSARR observations of the phase curves*, Icarus, Volume 452, March 2026 (arXiv:2603.08151).

2. Daniel V. Cotton, Kimberly Bott, Jeremy Bailey, Lucyna Kedziora-Chudczer, Ain De Horta, Graeme Melville, Normandy Filcek, Jonathan P. Marshall, Derek L. Buzasi, **Ievgeniia Boiko**, Nicholas Borsato, Jean Perkins, Daniela Opitz, Shannon Melrose, Gesa Grüning, Dag Evensberget and Jinglin Zhao, *Polarization position angle standard stars: a reassessment of θ and its variability for seventeen stars based on a decade of observations*, MNRAS, Volume 535, Issue 2, October 2024 (arXiv:2410.23466).
3. Daniel V. Cotton, Jeremy Bailey, Jean Perkins, Derek L. Buzasi, and **Ievgeniia Boiko**, *Deneb is a Large-amplitude Polarimetric Variable*, ApJL, Volume 967, Issue 2, April 2024 (arXiv:2404.17707).

In Preparation:

1. Daniel V. Cotton, Kimberly Bott, **Ievgeniia Boiko**, Lucyna Kedziora-Chudczer, Jeremy Bailey, Behrooz Karamiqucham and Jonathan Horner, *Towards an improved determination of the methane D/H ratio for Neptune*, in preparation for Icarus.

PRESENTATIONS & POSTERS

<i>American Astronomical Society 247 Winter Meeting</i> - Phoenix, AZ	January 2026
<i>Cal-Bridge Fall Conference</i> - UC Irvine	October 2025
<i>CNSM Student Research Symposium</i> - Cal. State Univ., Long Beach	September 2025
<i>Physics & Astronomy Dept. Colloquium</i> - Cal. State Univ., Long Beach	December 2024

COMPUTATIONAL SKILLS

Languages and Software:

Intermediate proficiency with Python, LaTeX
 Beginner proficiency with Fortran, HTML

Operating Systems:

Linux, MacOS

PROFESSIONAL ORGANIZATIONS

<i>Cal-Bridge Scholar Program</i> PHYSICS & ASTRONOMY UNDERGRADUATE SCHOLAR	2024 - 2026
<i>Society of Physics Students</i> - Cal. State Univ., Long Beach TREASURER (August 2024 - May 2025)	2023 - 2025
<i>Women in Physics</i> - Cal. State Univ., Long Beach PUBLICIST (August 2024 - May 2025)	2023 - 2025
<i>Astronomy Club</i> - Cal. State Univ., Long Beach SECRETARY (August 2024 - May 2025)	2023 - 2025

OUTREACH

Mobile Planetarium Program - Cal. State Univ., Long Beach

Fall 2024/Fall 2025

ASSISTANT

- Curated scripts and talks to be shown using the mobile planetarium
- Traveled to local high schools to assist with astronomy talks using CSULB's mobile planetarium
- Collaborated with director Dr. Joel Zinn and local schools' faculty and staff to ensure visit success

HONORS & AWARDS

President's List - Cal. State Univ., Long Beach

Fall 2025

Google Summer Research Assistantship - Cal. State Univ., Long Beach

Summer 2025

President's List - Cal. State Univ., Long Beach

Spring 2025

Olaf & Mary Jane Anfinson Scholarship - Cal. State Univ., Long Beach

2024 - 2025

Dean's List - Cal. State Univ., Long Beach

Spring 2024